



PCF Certificate Rotation Estimate

- **Director:** 192.168.2.2
- **Generated:** 2026-06-11 04:32:06 UTC
- **Horizon:** all certificates (no expiry limit)
- **Topology source:** maestro tp
- **Update rules:** bosh manifest (per-IG max_in_flight/canaries)

Summary

Metric	Value
Expiring certs (all (no limit))	112
— leaf / CA	105 / 7
— require Digicert (operator-supplied)	2
Live VMs	7
One foundation-wide Apply	48m – 1h 30m
Estimated rotation time	2h 24m – 4h 30m

Expiring certificates

Certificate	Type	Expires
/services/tls_leaf	Leaf	2027-05-30T16:32:48Z
/bosh_dns_health_client_tls	Leaf	2027-05-30T16:32:51Z
/bosh_dns_health_server_tls	Leaf	2027-05-30T16:32:51Z
/dns_api_client_tls	Leaf	2027-05-30T16:32:51Z
/dns_api_server_tls	Leaf	2027-05-30T16:32:51Z
.director.system_metrics_certificate	Leaf	2028-05-29T03:16:30Z
.properties.credhub_ssl	Leaf	2028-05-29T03:16:31Z
.properties.director_agent_ssl	Leaf	2028-05-29T03:16:31Z
.properties.director_ssl	Leaf	2028-05-29T03:16:31Z
.properties.uaa_ssl	Leaf	2028-05-29T03:16:31Z
.properties.blobstore_certificate	Leaf	2028-05-29T03:16:32Z
.properties.director_metrics_server_certificate	Leaf	2028-05-29T03:16:32Z
.properties.director_metrics_server_client_certificate	Leaf	2028-05-29T03:16:32Z
.properties.nats_director_client_certificate	Leaf	2028-05-29T03:16:33Z
.properties.nats_health_monitor_client_certificate	Leaf	2028-05-29T03:16:33Z
.properties.nats_server_certificate	Leaf	2028-05-29T03:16:33Z
.properties.networking_poe_ssl_certs[0].certificate	Leaf	2028-06-09T04:14:31Z
.uaa.service_provider_key_credentials	Leaf	2028-06-09T04:15:13Z
.properties.auctioneer_client_cert	Leaf	2028-06-09T04:17:51Z
.properties.auctioneer_server_cert	Leaf	2028-06-09T04:17:52Z
.properties.bbs_client_cert	Leaf	2028-06-09T04:17:52Z
.properties.bbs_server_cert	Leaf	2028-06-09T04:17:52Z
.properties.cloud_controller_client_cert	Leaf	2028-06-09T04:17:53Z
.properties.cloud_controller_logcache_tls_cert	Leaf	2028-06-09T04:17:53Z
.properties.cloud_controller_mutual_cert	Leaf	2028-06-09T04:17:54Z
.properties.cloud_controller_public_tls_cert	Leaf	2028-06-09T04:17:54Z
.properties.credhub_tls	Leaf	2028-06-09T04:17:54Z

Certificate	Type	Expires
<code>.properties.file_server_cert</code>	Leaf	2028-06-09T04:17:54Z
<code>.properties.log_cache_auth_proxy_client_tls_cert</code>	Leaf	2028-06-09T04:17:55Z
<code>.properties.log_cache_server_tls_cert</code>	Leaf	2028-06-09T04:17:55Z
<code>.properties.log_cache_cf_auth_proxy_metrics_tls</code>	Leaf	2028-06-09T04:17:56Z
<code>.properties.log_cache_gateway_metrics_tls</code>	Leaf	2028-06-09T04:17:56Z
<code>.properties.log_cache_metrics_tls</code>	Leaf	2028-06-09T04:17:56Z
<code>.properties.forwarder_agent_metrics_tls</code>	Leaf	2028-06-09T04:17:57Z
<code>.properties.log_cache_syslog_server_cert</code>	Leaf	2028-06-09T04:17:57Z
<code>.properties.log_cache_syslog_server_metrics_tls</code>	Leaf	2028-06-09T04:17:57Z
<code>.properties.log_cache_gateway_auth_proxy_tls_cert</code>	Leaf	2028-06-09T04:17:58Z
<code>.properties.loggregator_agent_metrics_tls</code>	Leaf	2028-06-09T04:17:58Z
<code>.properties.loggr_syslog_binding_cache_metrics_tls</code>	Leaf	2028-06-09T04:17:58Z
<code>.properties.loggr_udp_forwarder_tls</code>	Leaf	2028-06-09T04:17:58Z
<code>.properties.leadership_election_metrics_tls</code>	Leaf	2028-06-09T04:17:59Z
<code>.properties.loggr_metric_scraper_metrics_tls</code>	Leaf	2028-06-09T04:17:59Z
<code>.properties.leadership_election_tls</code>	Leaf	2028-06-09T04:18:00Z
<code>.properties.loggregator_client_cert</code>	Leaf	2028-06-09T04:18:00Z
<code>.properties.metric_registrar_orchestrator_tls</code>	Leaf	2028-06-09T04:18:00Z
<code>.properties.metric_registrar_rlp_tls</code>	Leaf	2028-06-09T04:18:00Z
<code>.properties.metric_registrar_endpoint_worker_scrape_tls</code>	Leaf	2028-06-09T04:18:01Z
<code>.properties.metric_registrar_secure_worker_tls</code>	Leaf	2028-06-09T04:18:01Z
<code>.properties.metric_registrar_worker_tls</code>	Leaf	2028-06-09T04:18:01Z
<code>.properties.metric_registrar_binding_cache_mtls</code>	Leaf	2028-06-09T04:18:02Z
<code>.properties.otel_collector_tls_cert</code>	Leaf	2028-06-09T04:18:02Z
<code>.properties.cf_hub_collector_tls_cert</code>	Leaf	2028-06-09T04:18:03Z
<code>.properties.nats_client_cert</code>	Leaf	2028-06-09T04:18:03Z
<code>.properties.nats_tls_external_cert</code>	Leaf	2028-06-09T04:18:04Z
<code>.properties.nats_tls_internal_cert</code>	Leaf	2028-06-09T04:18:04Z

Certificate	Type	Expires
<code>.properties.network_policy_agent_cert</code>	Leaf	2028-06-09T04:18:04Z
<code>.properties.network_policy_server_cert</code>	Leaf	2028-06-09T04:18:05Z
<code>.properties.network_policy_server_external_cert</code>	Leaf	2028-06-09T04:18:06Z
<code>.properties.prom_scraper_metrics_tls</code>	Leaf	2028-06-09T04:18:06Z
<code>.properties.prom_scraper_scrape_tls</code>	Leaf	2028-06-09T04:18:06Z
<code>.properties.pxc_internal_certificate</code>	Leaf	2028-06-09T04:18:06Z
<code>.properties.mysql_diag_agent_tls</code>	Leaf	2028-06-09T04:18:07Z
<code>.properties.mysql_galera_agent_tls</code>	Leaf	2028-06-09T04:18:07Z
<code>.properties.mysql_proxy_tls</code>	Leaf	2028-06-09T04:18:07Z
<code>.properties.pxc_server_certificate</code>	Leaf	2028-06-09T04:18:07Z
<code>.properties.mysql_replication_canary_tls</code>	Leaf	2028-06-09T04:18:08Z
<code>.properties.rep_client_cert</code>	Leaf	2028-06-09T04:18:08Z
<code>.properties.rep_server_cert_v2</code>	Leaf	2028-06-09T04:18:08Z
<code>.properties.reverse_log_proxy_gateway_client_tls_cc_cert</code>	Leaf	2028-06-09T04:18:09Z
<code>.properties.reverse_log_proxy_gateway_server_tls_cert</code>	Leaf	2028-06-09T04:18:09Z
<code>.properties.reverse_log_proxy_gateway_client_tls_rlp_cert</code>	Leaf	2028-06-09T04:18:10Z
<code>.properties.rlp_gateway_metrics_tls</code>	Leaf	2028-06-09T04:18:10Z
<code>.properties.routing_api_client_cert</code>	Leaf	2028-06-09T04:18:10Z
<code>.properties.routing_api_server_cert</code>	Leaf	2028-06-09T04:18:11Z
<code>.properties.routing_backends_client_cert_with_san</code>	Leaf	2028-06-09T04:18:11Z
<code>.properties.routing_lb_health_cert</code>	Leaf	2028-06-09T04:18:11Z
<code>.properties.service_discovery_client_tls</code>	Leaf	2028-06-09T04:18:11Z
<code>.properties.service_discovery_server_tls</code>	Leaf	2028-06-09T04:18:12Z
<code>.properties.ssh_proxy_backends_tls</code>	Leaf	2028-06-09T04:18:12Z
<code>.properties.syslog_agent_log_cache_tls</code>	Leaf	2028-06-09T04:18:12Z
<code>.properties.syslog_agent_metrics_tls</code>	Leaf	2028-06-09T04:18:12Z
<code>.properties.system_blobstore_public_endpoint_cert</code>	Leaf	2028-06-09T04:18:12Z
<code>.properties.tcp_routing_lb_health_cert</code>	Leaf	2028-06-09T04:18:13Z

Certificate	Type	Expires
.properties.tps_client_cert	Leaf	2028-06-09T04:18:13Z
.properties.syslog_agent_api_tls	Leaf	2028-06-09T04:18:14Z
.properties.system_metrics_tls_scraper_cert	Leaf	2028-06-09T04:18:14Z
.properties.trafficcontroller_tls_cert	Leaf	2028-06-09T04:18:14Z
.properties.binding_cache_api_tls	Leaf	2028-06-09T04:18:15Z
.properties.policy_server_asg_syncer	Leaf	2028-06-09T04:18:15Z
.nfs_server.blobstore_cert	Leaf	2028-06-09T04:18:16Z
.properties.cloud_controller_prom_scraper_scrape_tls	Leaf	2028-06-09T04:18:16Z
.properties.policy_server_asg_syncer_cc_client	Leaf	2028-06-09T04:18:16Z
.diego_database.locket_server_cert	Leaf	2028-06-09T04:18:17Z
.diego_database.silk_controller_server_cert	Leaf	2028-06-09T04:18:17Z
.diego_database.silk_daemon_client_cert	Leaf	2028-06-09T04:18:17Z
.cloud_controller.locket_client_cert	Leaf	2028-06-09T04:18:18Z
.uaa.ssl_credentials	Leaf	2028-06-09T04:18:18Z
.diego_brain.cc_uploader_server_cert	Leaf	2028-06-09T04:18:19Z
.loggregator_trafficcontroller.cc_trafficcontroller_tls_cert	Leaf	2028-06-09T04:18:19Z
.loggregator_trafficcontroller.rlp_tls_cert	Leaf	2028-06-09T04:18:20Z
.loggregator_trafficcontroller.statsdinjector_tls_cert	Leaf	2028-06-09T04:18:20Z
.loggregator_trafficcontroller.trafficcontroller_tls_cert	Leaf	2028-06-09T04:18:20Z
.doppler.doppler_tls_cert	Leaf	2028-06-09T04:18:21Z
.doppler.metron_tls_cert	Leaf	2028-06-09T04:18:21Z
/p-bosh/cf-dbe1a7580979a87638e7/diego-instance-identity-intermediate-ca-2-7	CA	2028-06-09T04:19:47Z
/cf/diego-instance-identity-root-ca-2-6	CA	2029-05-29T16:32:45Z
opsman-root-ca:3506e333	CA	2030-05-29T03:06:57Z
.properties.nats_client_ca.3506e33356bef86af963	CA	2030-05-29T03:06:57Z
.properties.root_ca.3506e33356bef86af963	CA	2030-05-29T03:06:57Z
/opsmgr/bosh_dns/tls_ca	CA	2030-05-29T16:32:50Z
/opsmgr/bosh_dns/san_migrated	Leaf	2030-05-29T16:32:52Z

Certificate	Type	Expires
/services/tls_ca	CA	2031-05-29T03:24:34Z

Foundation inventory

Deployment	VMs	Est. cert-rotation time
cf-dbe1a7580979a87638e7	7	2h 24m – 4h 30m

2 instance group(s) have `serial:false` — they may update in parallel, so real time can come in under the (conservative serial) estimate.

CA rotations

Certificate	Rotation	Expires
.properties.nats_client_ca.3506e33356bef86af963	3× foundation-wide	2030-05-29T03:06:57Z
.properties.root_ca.3506e33356bef86af963	3× foundation-wide	2030-05-29T03:06:57Z
/p-bosh/cf-dbe1a7580979a87638e7/diego-instance-identity-intermediate-ca-2-7	3× on deployment	2028-06-09T04:19:47Z
/opsmgr/bosh_dns/tls_ca	3× foundation-wide	2030-05-29T16:32:50Z
/cf/diego-instance-identity-root-ca-2-6	3× on deployment	2029-05-29T16:32:45Z
/services/tls_ca	2× foundation + 1× deployment	2031-05-29T03:24:34Z
opsman-root-ca:3506e333	3× foundation-wide	2030-05-29T03:06:57Z

Leaf certificates

Deployment	Leaf certs
p-bosh (not VM-counted)	11
cf-dbe1a7580979a87638e7	94

Operator-supplied certificates — require Digicert

Not auto-generated. A new certificate must be obtained from Digicert before rotation (out-of-band; not included in the Apply Changes time above).

Certificate	Expires
.properties.networking_poe_ssl_certs[0].certificate	2028-06-09T04:14:31Z
.uaa.service_provider_key_credentials	2028-06-09T04:15:13Z

Estimate breakdown

Campaign	Applies	Time (low – high)
Leaf certs	folded into the shared apply #2	included
CA certs (full batch)	3× foundation-wide on all tiles — all CAs + leaves	2h 24m – 4h 30m
Total		2h 24m – 4h 30m

Model & assumptions

- 20m overhead per Apply Changes; 4–10m per VM.
- A FOUNDATION CA = 3 foundation-wide applies; a services TLS CA = 2 foundation + 1 deployment applies; a BOSH trusted-store CA = 1 foundation-wide apply (swap in place); a deployment CA = 3 on its deployment.
- Estimate is **Apply Changes compute time only** — it excludes change-window / approval gaps between phases, which often dominate the wall-clock for CA rotations.